

Smart Parking Platform

Roles & Resources, Resource & Cost Plan, and RACI Matrix

Software Corp | Project Management Homework

Planning Assumptions

- The project runs six months, and each full-time month equals 160 working hours.
- All names are fictional, and hourly rates reflect assumed Houston market pay for each skill level.
- Cost estimates use the bottom-up method, building each total from hours, rates, and unit prices.
- A ten percent contingency reserve covers unplanned scope, rework, and price changes.
- Hardware, software, and service costs come from assumed vendor pricing, not quotes.

1. Roles & Resources

Table 1 lists each project role with a fictional name, skill level, and allocation. Two roles were added to the original list: Business Analyst and Security & Compliance Specialist.

Table 1: Project roles, assumed skill levels, and allocations

Role	Name	Skill Level	Allocation	Active Months	Main Responsibility
Project Manager	Maya Okafor	Senior	50%	Months 1-6	Runs schedule, budget, risks, and status reporting
Business Analyst	Elena Petrova	Mid	50%	Months 1-4	Gathers requirements and maintains the SRS
UX/UI Specialist	Priya Nair	Mid	50%	Months 1-5	Designs user flows, wireframes, and app screens
iOS Developer	Daniel Reyes	Senior	100%	Months 2-6	Builds and ships the iPhone driver app
Android Developer	Hannah Brooks	Mid	100%	Months 2-6	Builds and ships the Android driver app
Frontend Developer	Marcus Lee	Mid	75%	Months 2-5	Builds the web dashboard for operators and admins
Backend Developer	Tessa Alvarez	Senior	100%	Months 2-6	Builds booking, availability, and account services
Cloud Engineer	Omar Haddad	Senior	50%	Months 1-6	Sets up hosting, pipelines, and monitoring
Embedded Systems Engineer	Wei Zhang	Senior	75%	Months 1-5	Develops sensor hardware and firmware
Security & Compliance Specialist	Victor Hughes	Senior	25%	Months 3-5	Reviews security, privacy, and payment compliance
QA Engineer	Grace Kim	Senior	75%	Months 3-6	Owns test strategy, automation, and release sign-off
Tester 1	Jordan Pierce	Junior	50%	Months 4-6	Runs manual test cases and logs defects
Tester 2	Alicia Torres	Junior	50%	Months 4-6	Runs manual test cases and logs defects
Data Collector 1	Carlos Mendez	Junior	25%	Months 1-3	Records lot, pricing, and occupancy data in Houston
Data Collector 2	Naomi Johnson	Junior	25%	Months 1-3	Records lot, pricing, and occupancy data in Houston

Role	Name	Skill Level	Allocation	Active Months	Main Responsibility
Data Collector 3	Tyler Nguyen	Junior	25%	Months 1-3	Records lot, pricing, and occupancy data in Houston
Data Collector 4	Imani Brooks	Junior	25%	Months 1-3	Records lot, pricing, and occupancy data in Houston
Data Collector 5	Sam Delgado	Junior	25%	Months 1-3	Records lot, pricing, and occupancy data in Houston
Data Analyst	Nina Patel	Mid	50%	Months 2-6	Turns field data into demand and pricing insights
Finance & Payments Specialist	Robert Chen	Senior	25%	Months 2-5	Designs billing, payouts, and payment gateway rules
B2B Partner Support	Sofia Marino	Mid	50%	Months 4-6	Onboards and supports lot operators and city partners
Customer Support Specialist 1	Liam O'Connor	Junior	50%	Months 5-6	Handles driver tickets, refunds, and disputes
Customer Support Specialist 2	Aisha Rahman	Junior	50%	Months 5-6	Handles driver tickets, refunds, and disputes

2. Resource & Cost Plan

2.1 Resource Plan

Hours equal FTE percent times 160 hours times the months each person is active.

Table 2: Resource plan by person

Role	Name	FTE %	Estimated Hours	Hourly Rate	Cost
Project Manager	Maya Okafor	0.50	480	\$65	\$31,200
Business Analyst	Elena Petrova	0.50	320	\$55	\$17,600
UX/UI Specialist	Priya Nair	0.50	400	\$50	\$20,000
iOS Developer	Daniel Reyes	1.00	800	\$60	\$48,000
Android Developer	Hannah Brooks	1.00	800	\$55	\$44,000
Frontend Developer	Marcus Lee	0.75	480	\$50	\$24,000
Backend Developer	Tessa Alvarez	1.00	800	\$65	\$52,000
Cloud Engineer	Omar Haddad	0.50	480	\$70	\$33,600
Embedded Systems Engineer	Wei Zhang	0.75	600	\$65	\$39,000
Security & Compliance Specialist	Victor Hughes	0.25	120	\$75	\$9,000
QA Engineer	Grace Kim	0.75	480	\$50	\$24,000
Tester 1	Jordan Pierce	0.50	240	\$30	\$7,200
Tester 2	Alicia Torres	0.50	240	\$30	\$7,200
Data Collector 1	Carlos Mendez	0.25	120	\$22	\$2,640
Data Collector 2	Naomi Johnson	0.25	120	\$22	\$2,640
Data Collector 3	Tyler Nguyen	0.25	120	\$22	\$2,640
Data Collector 4	Imani Brooks	0.25	120	\$22	\$2,640
Data Collector 5	Sam Delgado	0.25	120	\$22	\$2,640

Role	Name	FTE %	Estimated Hours	Hourly Rate	Cost
Data Analyst	Nina Patel	0.50	400	\$48	\$19,200
Finance & Payments Specialist	Robert Chen	0.25	160	\$60	\$9,600
B2B Partner Support	Sofia Marino	0.50	240	\$38	\$9,120
Customer Support Specialist 1	Liam O'Connor	0.50	160	\$25	\$4,000
Customer Support Specialist 2	Aisha Rahman	0.50	160	\$25	\$4,000
TOTAL			7,960		\$415,920

2.2 Cost Plan (Bottom-Up)

Labor is grouped by team, then non-labor items are added line by line.

Table 3: Bottom-up cost plan

Item	Category	Qty/Months	Unit Cost	Duration (months)	Total	Notes
Management & analysis labor	Labor	1,360 hours	\$57.06/hour avg	6	\$77,600	PM, BA, analyst, finance
Design & engineering labor	Labor	4,480 hours	\$60.18/hour avg	6	\$269,600	Design, apps, backend, cloud, embedded, security
QA & testing labor	Labor	960 hours	\$40.00/hour avg	4	\$38,400	QA engineer and 2 testers
Field data collection labor	Labor	600 hours	\$22.00/hour avg	3	\$13,200	5 collectors, Houston lots
Support & partner labor	Labor	560 hours	\$30.57/hour avg	3	\$17,120	B2B and 2 support specialists
Cloud hosting	Infra	\$450/month	\$450	6	\$2,700	Dev, staging, and production
Maps & geocoding API	Infra	\$250/month	\$250	5	\$1,250	Live lot maps and routing
Support helpdesk tool	Tools	3 seats	\$30/seat/mo	2	\$180	Ticketing for driver issues
Dev & design licenses	Tools	one-off	\$1,800	N/A	\$1,800	GitHub, Jira, Figma, Sentry
App store accounts	Tools	one-off	\$124	N/A	\$124	Apple \$99 and Google \$25
IoT parking sensors	Hardware	60 units	\$95/unit	N/A	\$5,700	Pilot lots in Houston
Sensor gateways	Hardware	6 units	\$180/unit	N/A	\$1,080	One per pilot lot
Test devices	Hardware	8 phones	\$300/unit	N/A	\$2,400	Mixed iOS and Android models
Field data equipment & mileage	Field	5 collectors	\$150/month	3	\$2,250	Gas, parking, and phone data
PCI DSS assessment	Compliance	one-off	\$4,500	N/A	\$4,500	Payment security review
Contingency	Reserve	10%	applied to subtotal	N/A	\$43,790	10% of \$437,904
TOTAL PROJECT COST					\$481,694	

2.3 Monthly Budgeted Cashflow

Labor is paid in the months each person works, and purchases land when needed.

Table 4: Monthly cashflow over six months

Cost Line	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Total
Management & analysis labor	\$9,600	\$15,840	\$15,840	\$15,840	\$11,440	\$9,040	\$77,600
Design & engineering labor	\$17,400	\$52,200	\$55,200	\$55,200	\$55,200	\$34,400	\$269,600
QA & testing labor	-	-	\$6,000	\$10,800	\$10,800	\$10,800	\$38,400
Field data collection labor	\$4,400	\$4,400	\$4,400	-	-	-	\$13,200
Support & partner labor	-	-	-	\$3,040	\$7,040	\$7,040	\$17,120
Cloud, maps & helpdesk	\$450	\$700	\$700	\$700	\$790	\$790	\$4,130
Licenses & accounts	\$1,800	-	-	\$124	-	-	\$1,924
Hardware & field equipment	\$750	\$7,530	\$3,150	-	-	-	\$11,430
PCI compliance	-	-	-	-	\$4,500	-	\$4,500
Subtotal	\$34,400	\$80,670	\$85,290	\$85,704	\$89,770	\$62,070	\$437,904
Contingency (10%)	\$3,440	\$8,067	\$8,529	\$8,570	\$8,977	\$6,207	\$43,790
Monthly cash out	\$37,840	\$88,737	\$93,819	\$94,274	\$98,747	\$68,277	\$481,694
Cumulative cash out	\$37,840	\$126,577	\$220,396	\$314,670	\$413,417	\$481,694	

Total budget: \$481,694 including a \$43,790 contingency reserve.

3. RACI Matrix

Table 5 shows who is responsible, accountable, consulted, and informed for each activity.

R = Responsible (does the work). A = Accountable (one owner, final sign-off). C = Consulted (gives input). I = Informed (kept updated). A/R = same person owns and does the work.

Activity	PM	BA	UX	iOS	AND	FE	BE	CLD	EMB	SEC	QA	TST	DC	DA	FIN	B2B	CS
1. Project plan, schedule, and budget	A/R	C	I	I	I	I	C	C	C	I	C	I	I	I	C	I	I
2. Requirements gathering and SRS	A	R	C	C	C	C	C	C	C	C	C	I	I	C	C	C	C
3. Houston parking data collection	A	C	I	I	I	I	I	I	C	I	I	I	R	R	I	I	I
4. Data analysis and demand reporting	A	C	C	I	I	I	C	I	I	I	I	I	C	R	C	I	I
5. UX research and UI design	A	C	R	C	C	C	I	I	I	I	C	I	I	C	I	C	C
6. System architecture and API design	C	C	I	C	C	C	A/R	R	C	C	I	I	I	C	C	C	I
7. Cloud setup and deployment pipeline	I	I	I	I	I	I	C	A/R	I	C	C	I	I	I	I	I	I
8. Sensor hardware and firmware	C	C	I	I	I	I	C	C	A/R	C	C	I	C	I	I	I	I
9. Backend services (booking, availability, accounts)	I	C	I	C	C	C	A/R	C	C	C	C	I	I	I	C	C	I
10. iOS app development	I	C	C	A/R	I	I	C	I	I	C	C	I	I	I	I	I	I
11. Android app development	I	C	C	I	A/R	I	C	I	I	C	C	I	I	I	I	I	I
12. Web dashboard for operators and admins	I	C	C	I	I	A/R	C	I	I	C	C	I	I	C	C	C	I
13. Payments integration and billing	I	C	C	C	C	C	R	I	I	C	C	I	I	I	A	C	I
14. Security and compliance review	I	C	I	I	I	I	C	C	C	A/R	C	I	I	I	C	I	I
15. Test planning and test cases	I	C	I	I	I	I	I	I	I	I	A/R	R	I	I	I	I	I
16. System and user acceptance testing	I	C	C	C	C	C	C	C	C	I	A	R	I	I	C	C	C
17. Defect fixing and release readiness	A	I	C	R	R	R	R	C	R	C	R	C	I	I	I	I	I
18. B2B operator onboarding and partner support	C	C	I	I	I	I	C	I	C	I	I	I	I	I	C	A/R	C
19. Customer support setup and training	A	C	C	I	I	I	I	I	I	I	C	I	I	I	C	C	R
20. Pilot launch in Houston	A	I	I	C	C	C	C	R	R	C	C	I	I	I	C	R	R
21. Status reporting and risk management	A/R	C	I	I	I	I	C	C	C	C	C	I	I	I	C	I	I
22. Post-launch monitoring and performance reporting	A	I	I	I	I	I	C	R	C	I	C	I	I	R	C	C	C

Role abbreviations used in the matrix

PM	Project Manager	BE	Backend Developer	DC	Data Collectors 1 to 5
BA	Business Analyst	CLD	Cloud Engineer	DA	Data Analyst
UX	UX/UI Specialist	EMB	Embedded Systems Engineer	FIN	Finance & Payments Specialist

iOS	iOS Developer	SEC	Security & Compliance Specialist	B2B	B2B Partner Support
AND	Android Developer	QA	QA Engineer	CS	Customer Support Specialists 1 and 2
FE	Frontend Developer	TST	Testers 1 and 2		